# DDA3020 Machine Learning

Tutorial: Linear Regression

April 2026

## Instructions

This tutorial covers the core concepts of Linear Regression, including deterministic and probabilistic perspectives, analytical solutions, gradient descent, multiple outputs, classification extensions, and variants like Ridge, Lasso, Polynomial, and Robust regression. Show your working clearly for all calculation and derivation questions.

## Useful Formulas

- Linear Hypothesis Function: $f_{\mathbf{w}}(\mathbf{x}) = \mathbf{w}^\top \mathbf{x}$ (where $\mathbf{x}$ is augmented with 1 for bias)
- Cost Function (MSE): $J(\mathbf{w}) = \frac{1}{2m} \sum_{i=1}^{m} (\mathbf{w}^\top \mathbf{x}_i - y_i)^2 = \frac{1}{2m} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2$
- Analytical Solution (Normal Equation): $\mathbf{w}^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$
- Gradient Descent Update: $\mathbf{w} \leftarrow \mathbf{w} - \alpha \mathbf{X}^\top (\mathbf{X}\mathbf{w} - \mathbf{y})$
- Ridge Regression Solution: $\mathbf{w}^* = (\mathbf{X}^\top \mathbf{X} + \lambda \hat{\mathbf{I}}_d)^{-1} \mathbf{X}^\top \mathbf{y}$
- Probabilistic Model: $y = \mathbf{w}^\top \mathbf{x} + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma^2)$
- Maximum Likelihood Estimation (MLE): $\mathbf{w}_{MLE} = \arg\max_{\mathbf{w}} \log \prod_{i=1}^{m} p(y_i | \mathbf{x}_i, \mathbf{w})$
- Maximum A Posteriori (MAP): $\mathbf{w}_{MAP} = \arg\max_{\mathbf{w}} \left[\sum_{i=1}^{m} \log p(y_i | \mathbf{x}_i, \mathbf{w}) + \log p(\mathbf{w})\right]$

# Exercises

## Multiple Choice Questions

Q1. In linear regression, the analytical solution $\mathbf{w}^* = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$ requires the matrix $\mathbf{X}^\top\mathbf{X}$ to be invertible. Which of the following methods guarantees an invertible matrix by adding a regularization term?

(A) Lasso Regression

(B) Ridge Regression

(C) Polynomial Regression

(D) Robust Linear Regression

Q2. From a probabilistic perspective, standard least squares linear regression is equivalent to Maximum Likelihood Estimation (MLE) under which assumption about the residual errors?

(A) They follow a Uniform distribution

(B) They follow a Laplace distribution

(C) They follow a Gaussian (Normal) distribution

(D) They follow a Student's t-distribution

Q3. Which of the following statements comparing the closed-form solution and gradient descent for linear regression is INCORRECT?

(A) The closed-form solution does not require choosing a learning rate hyper-parameter.

(B) Gradient descent is generally faster than the closed-form solution when the number of features $d$ is very large.

(C) The closed-form solution has a computational complexity of $O(d^3 + md^2)$.

(D) Gradient descent always finds the global optimum in a single iteration for linear regression.

Q4. In Ridge regression, the parameter $\mathbf{w}$ is estimated using Maximum A Posteriori (MAP). What prior distribution is assumed for $\mathbf{w}$?

(A) Uniform prior

(B) Gaussian prior

(C) Laplace prior

(D) Exponential prior

Q5. Robust linear regression uses the $\ell_1$ loss function instead of the $\ell_2$ (squared error) loss. What is the primary advantage of this approach?

(A) It makes the objective function strictly convex and continuously differentiable everywhere.

(B) It guarantees a closed-form analytical solution.

(C) It is less sensitive to outliers in the training data.

(D) It automatically performs feature selection by shrinking some weights to exactly zero.

## Calculation Questions

Q6. Consider a simple 1D linear regression problem with the following dataset of $m = 3$ points: $(x_1, y_1) = (1, 2)$, $(x_2, y_2) = (2, 4)$, $(x_3, y_3) = (3, 5)$. We want to fit a linear model $f_w(x) = w_0 + w_1 x$.

(1) Construct the design matrix $\mathbf{X}$ and the target vector $\mathbf{y}$.

(2) Calculate $\mathbf{X}^\top\mathbf{X}$ and $\mathbf{X}^\top\mathbf{y}$.

(3) Find the analytical solution $\mathbf{w}^* = (w_0, w_1)^\top$.

(4) Predict the value of $y$ for a new input $x_{new} = 4$.

Q7. Suppose we apply Ridge regression to the same dataset from Q6, with a regularization parameter $\lambda = 2$.

(1) Write down the modified matrix $(\mathbf{X}^\top\mathbf{X} + \lambda\hat{\mathbf{I}}_d)$, where $\hat{\mathbf{I}}_d$ is the identity matrix with the top-left element set to 0 (to avoid penalizing the bias term $w_0$).

(2) Calculate the new Ridge regression solution $\mathbf{w}^*_{ridge}$.

## Short Answer Questions

Q8. Explain the motivation behind Polynomial Regression. How does it allow a linear model to fit non-linear data?

Q9. Compare Ridge Regression and Lasso Regression in terms of their regularization penalty and their effect on the learned model parameters.

## Derivation Questions

Q10. Derive the analytical solution for standard linear regression. Starting from the matrix form of the squared error cost function $J(\mathbf{w}) = (\mathbf{X}\mathbf{w} - \mathbf{y})^\top(\mathbf{X}\mathbf{w} - \mathbf{y})$, show the steps to find the optimal weight vector $\mathbf{w}^*$.

Q11. Show that Maximum A Posteriori (MAP) estimation with a zero-mean Gaussian prior on the weights $\mathbf{w}$ is equivalent to Ridge Regression. Assume the likelihood is $p(y_i|\mathbf{x}_i, \mathbf{w}) = \mathcal{N}(y_i|\mathbf{w}^\top\mathbf{x}_i, \sigma^2)$ and the prior is $p(\mathbf{w}) = \mathcal{N}(\mathbf{w}|\mathbf{0}, \tau^2\mathbf{I})$.

# Solutions

## Multiple Choice Questions

Q1. **B**. Ridge regression adds $\lambda\hat{\mathbf{I}}_d$ to $\mathbf{X}^\top\mathbf{X}$, which guarantees the resulting matrix is invertible for $\lambda > 0$.

Q2. **C**. Standard least squares is equivalent to MLE under the assumption that the residual errors follow a zero-mean Gaussian distribution.

Q3. **D**. Gradient descent requires multiple iterations to converge to the optimum; it does not find it in a single iteration.

Q4. **B**. Ridge regression corresponds to MAP estimation with a Gaussian prior on the weights. Lasso corresponds to a Laplace prior.

Q5. **C**. The $\ell_1$ loss grows linearly with the residual, whereas $\ell_2$ loss grows quadratically. Thus, $\ell_1$ loss (Robust regression) is much less sensitive to large outliers.

## Calculation Questions

Q6. (1) The augmented design matrix $\mathbf{X}$ and target vector $\mathbf{y}$ are:

$$\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}$$

(2)

$$\mathbf{X}^\top\mathbf{X} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\mathbf{X}^\top\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 2+4+5 \\ 2+8+15 \end{bmatrix} = \begin{bmatrix} 11 \\ 25 \end{bmatrix}$$

(3) We need to solve $(\mathbf{X}^\top\mathbf{X})\mathbf{w} = \mathbf{X}^\top\mathbf{y}$, or find the inverse of $\mathbf{X}^\top\mathbf{X}$.

$$(\mathbf{X}^\top\mathbf{X})^{-1} = \frac{1}{3(14) - 6(6)} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix} = \frac{1}{42 - 36} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}$$

$$\mathbf{w}^* = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix} \begin{bmatrix} 11 \\ 25 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 14(11) - 6(25) \\ -6(11) + 3(25) \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 154 - 150 \\ -66 + 75 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 4 \\ 9 \end{bmatrix} = \begin{bmatrix} 2/3 \\ 3/2 \end{bmatrix}$$

So $w_0 = 2/3 \approx 0.67$ and $w_1 = 3/2 = 1.5$. The model is $y = 2/3 + 1.5x$.

(4) For $x_{new} = 4$, $y_{new} = 2/3 + 1.5(4) = 2/3 + 6 = 20/3 \approx 6.67$.

Q7. (1) The regularization matrix is $\lambda \hat{\mathbf{I}}_d = 2 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$.

$$\mathbf{X}^\top \mathbf{X} + \lambda \hat{\mathbf{I}}_d = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 16 \end{bmatrix}$$

(2) We need to solve $(\mathbf{X}^\top \mathbf{X} + \lambda \hat{\mathbf{I}}_d)\mathbf{w} = \mathbf{X}^\top \mathbf{y}$.

$$\begin{bmatrix} 3 & 6 \\ 6 & 16 \end{bmatrix}^{-1} = \frac{1}{3(16) - 6(6)} \begin{bmatrix} 16 & -6 \\ -6 & 3 \end{bmatrix} = \frac{1}{48 - 36} \begin{bmatrix} 16 & -6 \\ -6 & 3 \end{bmatrix} = \frac{1}{12} \begin{bmatrix} 16 & -6 \\ -6 & 3 \end{bmatrix}$$

$$\mathbf{w}^*_{ridge} = \frac{1}{12} \begin{bmatrix} 16 & -6 \\ -6 & 3 \end{bmatrix} \begin{bmatrix} 11 \\ 25 \end{bmatrix} = \frac{1}{12} \begin{bmatrix} 16(11) - 6(25) \\ -6(11) + 3(25) \end{bmatrix} = \frac{1}{12} \begin{bmatrix} 176 - 150 \\ -66 + 75 \end{bmatrix} = \frac{1}{12} \begin{bmatrix} 26 \\ 9 \end{bmatrix} = \begin{bmatrix} 13/6 \\ 3/4 \end{bmatrix}$$

So $w_0 = 13/6 \approx 2.17$ and $w_1 = 3/4 = 0.75$. The model is $y = 13/6 + 0.75x$. Notice how the slope $w_1$ has been shrunk from 1.5 to 0.75 due to regularization.

## Short Answer Questions

Q8. Polynomial regression is motivated by the fact that many real-world datasets are not linearly separable or cannot be well-approximated by a simple straight line or hyperplane. It allows a linear model to fit non-linear data by applying a basis expansion $\phi(\mathbf{x})$ to the original features. For example, it maps original features $x_1, x_2$ to a higher-dimensional space including polynomial terms like $x_1^2, x_2^2, x_1 x_2$. The model $f_{\mathbf{w}}(\mathbf{x}) = \mathbf{w}^\top \phi(\mathbf{x})$ is non-linear with respect to the original input $\mathbf{x}$, but it remains a linear function with respect to the parameters $\mathbf{w}$. Thus, we can still use standard linear regression algorithms to find the optimal weights.

Q9. **Ridge Regression** uses an $\ell_2$ regularization penalty ($\lambda \|\mathbf{w}\|_2^2$). It shrinks all parameter values towards zero proportionally, which helps prevent overfitting and handles multicollinearity, but it rarely forces weights to be exactly zero.

**Lasso Regression** uses an $\ell_1$ regularization penalty ($\alpha \|\mathbf{w}\|_1$). It not only shrinks parameters but also tends to produce sparse solutions, meaning it forces the weights of less important features to exactly zero. This makes Lasso useful for automatic feature selection.

## Derivation Questions

Q10. **Step 1: Expand the cost function**

$$\begin{aligned} J(\mathbf{w}) &= (\mathbf{X}\mathbf{w} - \mathbf{y})^\top (\mathbf{X}\mathbf{w} - \mathbf{y}) \\ &= (\mathbf{w}^\top \mathbf{X}^\top - \mathbf{y}^\top)(\mathbf{X}\mathbf{w} - \mathbf{y}) \\ &= \mathbf{w}^\top \mathbf{X}^\top \mathbf{X}\mathbf{w} - \mathbf{w}^\top \mathbf{X}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{X}\mathbf{w} + \mathbf{y}^\top \mathbf{y} \end{aligned}$$

Since $\mathbf{w}^\top \mathbf{X}^\top \mathbf{y}$ is a scalar, it is equal to its transpose $\mathbf{y}^\top \mathbf{X}\mathbf{w}$. Thus:

$$J(\mathbf{w}) = \mathbf{w}^\top \mathbf{X}^\top \mathbf{X}\mathbf{w} - 2\mathbf{y}^\top \mathbf{X}\mathbf{w} + \mathbf{y}^\top \mathbf{y}$$

**Step 2: Take the derivative with respect to w** Using matrix calculus rules $\frac{\partial}{\partial \mathbf{w}}(\mathbf{w}^\top \mathbf{A}\mathbf{w}) = 2\mathbf{A}\mathbf{w}$ (for symmetric $\mathbf{A}$) and $\frac{\partial}{\partial \mathbf{w}}(\mathbf{b}^\top \mathbf{w}) = \mathbf{b}$:

$$\frac{\partial J(\mathbf{w})}{\partial \mathbf{w}} = 2\mathbf{X}^\top \mathbf{X}\mathbf{w} - 2\mathbf{X}^\top \mathbf{y}$$

**Step 3: Set the derivative to zero and solve for w**

$$2\mathbf{X}^\top \mathbf{X}\mathbf{w} - 2\mathbf{X}^\top \mathbf{y} = 0$$

$$\mathbf{X}^\top \mathbf{X}\mathbf{w} = \mathbf{X}^\top \mathbf{y}$$

Assuming $\mathbf{X}^\top \mathbf{X}$ is invertible, we multiply both sides by $(\mathbf{X}^\top \mathbf{X})^{-1}$:

$$\mathbf{w}^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Q11. **Step 1: Formulate the MAP objective** The MAP estimate maximizes the posterior probability:

$$\mathbf{w}_{MAP} = \arg\max_{\mathbf{w}} \left[ \sum_{i=1}^{m} \log p(y_i | \mathbf{x}_i, \mathbf{w}) + \log p(\mathbf{w}) \right]$$

**Step 2: Substitute the distributions** The log-likelihood for the Gaussian noise is:

$$\log p(y_i|\mathbf{x}_i, \mathbf{w}) = \log \left( \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left( -\frac{(y_i - \mathbf{w}^\top \mathbf{x}_i)^2}{2\sigma^2} \right) \right) = -\frac{1}{2\sigma^2}(y_i - \mathbf{w}^\top \mathbf{x}_i)^2 + C_1$$

The log-prior for the Gaussian weights is:

$$\log p(\mathbf{w}) = \log \left( \frac{1}{(2\pi\tau^2)^{d/2}} \exp\left( -\frac{\|\mathbf{w}\|_2^2}{2\tau^2} \right) \right) = -\frac{1}{2\tau^2}\|\mathbf{w}\|_2^2 + C_2$$

**Step 3: Simplify the objective** Substitute these into the MAP objective and drop the constants $C_1, C_2$:

$$\mathbf{w}_{MAP} = \arg\max_{\mathbf{w}} \left[ -\frac{1}{2\sigma^2} \sum_{i=1}^{m} (y_i - \mathbf{w}^\top \mathbf{x}_i)^2 - \frac{1}{2\tau^2} \|\mathbf{w}\|_2^2 \right]$$

Maximizing this is equivalent to minimizing its negative:

$$\mathbf{w}_{MAP} = \arg\min_{\mathbf{w}} \left[ \frac{1}{2\sigma^2} \sum_{i=1}^{m} (\mathbf{w}^\top \mathbf{x}_i - y_i)^2 + \frac{1}{2\tau^2} \|\mathbf{w}\|_2^2 \right]$$

**Step 4: Show equivalence to Ridge Regression** Multiply the entire objective by $2\sigma^2$ (which doesn't change the argmin):

$$\mathbf{w}_{MAP} = \arg\min_{\mathbf{w}} \left[ \sum_{i=1}^{m} (\mathbf{w}^\top \mathbf{x}_i - y_i)^2 + \frac{\sigma^2}{\tau^2} \|\mathbf{w}\|_2^2 \right]$$

Let $\lambda = \frac{\sigma^2}{\tau^2}$. The objective becomes:

$$\mathbf{w}_{MAP} = \arg\min_{\mathbf{w}} \left[ \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda \|\mathbf{w}\|_2^2 \right]$$

This is exactly the objective function for Ridge Regression.